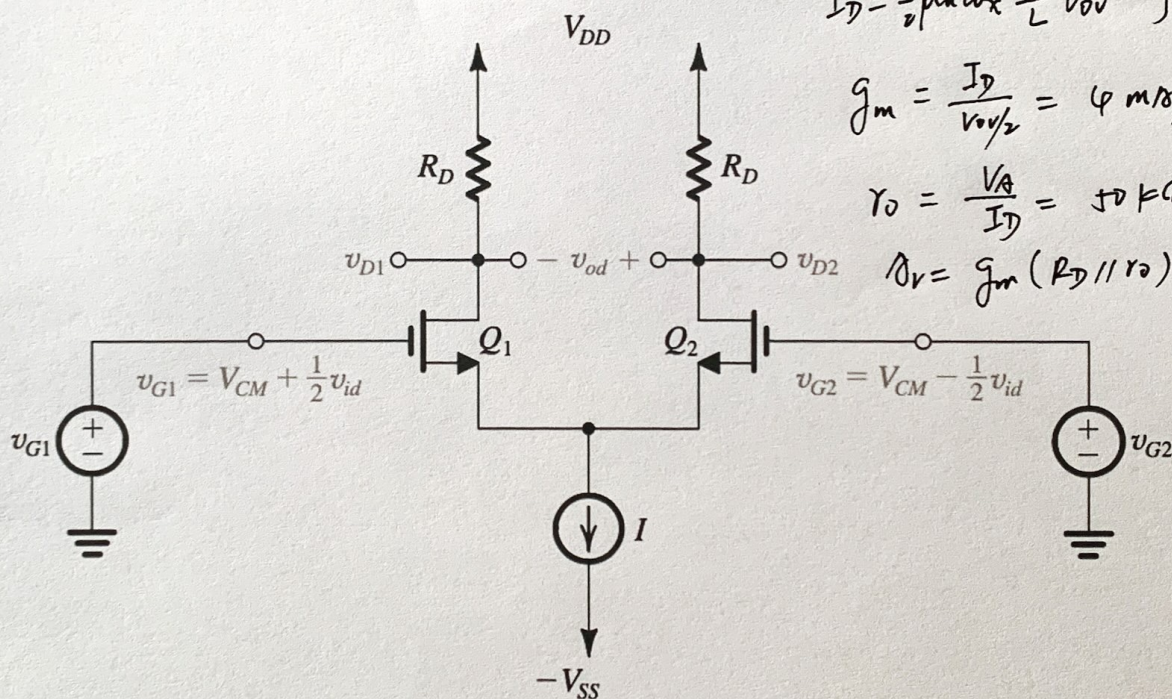


作业

9.4 A MOS differential amplifier is operated at a total current of 0.8 mA, using transistors with a W/L ratio of 100, $\mu_n C_{ox} = 0.2 \text{ mA/V}^2$, $V_A = 20 \text{ V}$, and $R_D = 5 \text{ k}\Omega$. Find V_{OV} , g_m , r_o , and A_d .

Ans. 0.2 V; 4 mA/V; 50 k Ω ; 18.2 V/V



$$\begin{aligned} I_D &= 0.4 \text{ mA} \\ I_D &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{OV}^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} I_D &= 0.4 \text{ mA} \\ I_D &= \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{OV}^2 \end{aligned}} \right\} \Rightarrow V_{OV} = 0.2 \text{ V}$$

$$g_m = \frac{I_D}{V_{OV}/2} = 4 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = 50 \text{ k}\Omega$$

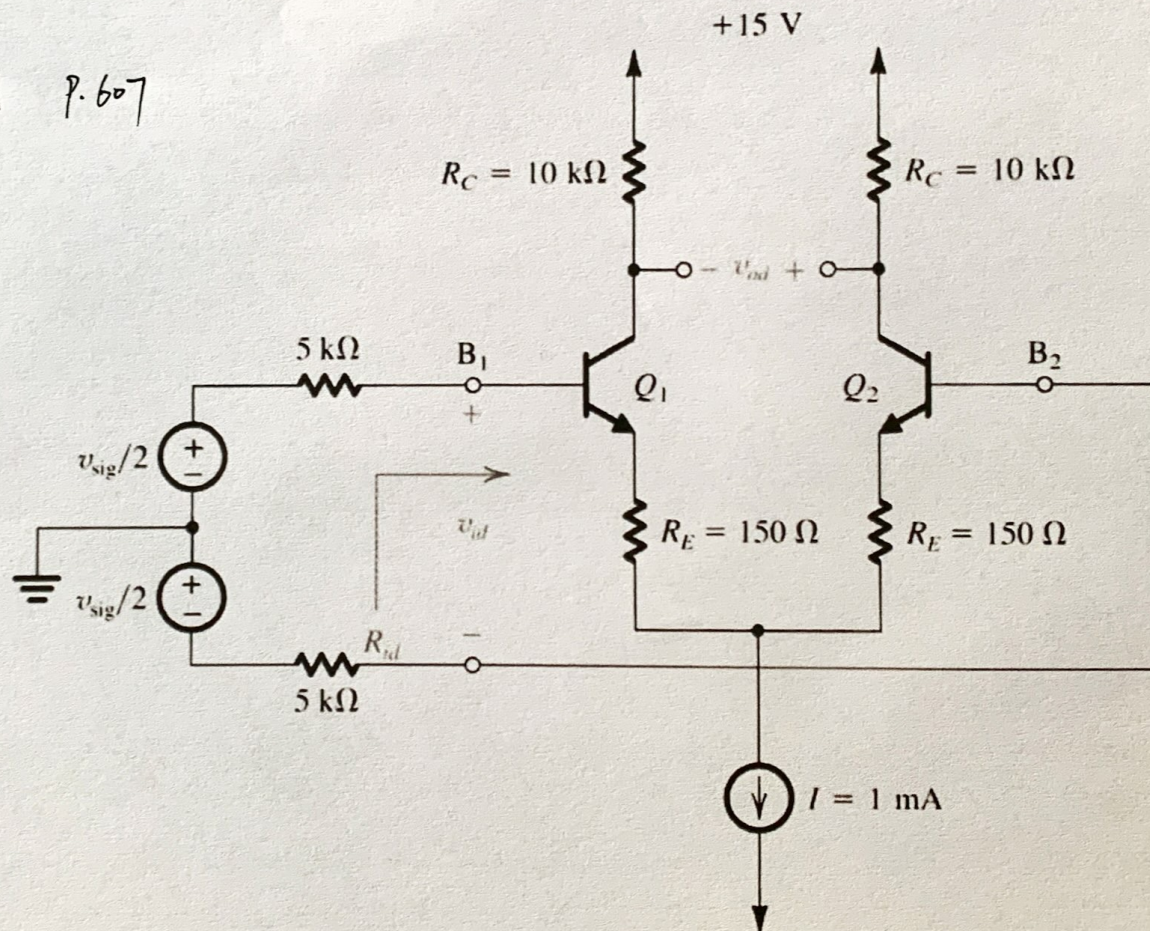
$$A_v = g_m (R_D \parallel r_o) = 18.2 \text{ V/V}$$

Example 9.3

The differential amplifier in Fig. 9.23 uses transistors with $\beta = 100$. Evaluate the following:

- (a) The input differential resistance R_{id} .
- (b) The overall differential voltage gain v_{od}/v_{sig} (neglect the effect of r_o).

见教材 p. 607



9.17 A current-mirror-loaded MOS differential amplifier of the type shown in Fig. 9.32(a) is specified as follows: $(W/L)_n = 100$, $(W/L)_p = 200$, $\mu_n C_{ox} = 2\mu_p C_{ox} = 0.2 \text{ mA/V}^2$, $V_{An} = |V_{Ap}| = 20 \text{ V}$, and $I = 0.8 \text{ mA}$. Calculate G_m , R_o , and A_d .

Ans. 4 mA/V ; $25 \text{ k}\Omega$; 100 V/V

$$I_D = 0.4 \text{ mA}$$

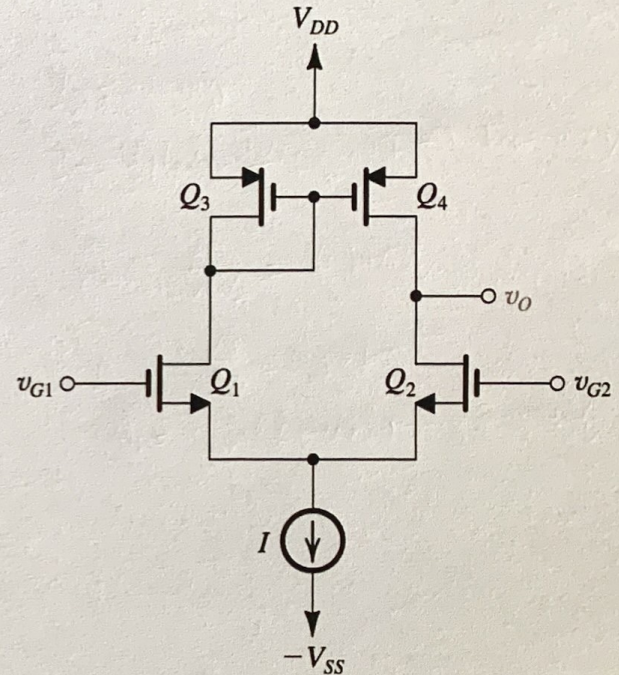
$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_n V_{ov}^2 \quad \left. \vphantom{I_D} \right\} \Rightarrow V_{ov} = 0.2 \text{ V}$$

$$G_m = g_{m,n} = \frac{I_D}{V_{ov}/2} = 4 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = 50 \text{ k}\Omega$$

$$R_o = \frac{1}{2} r_o = 25 \text{ k}\Omega$$

$$A_d = G_m R_o = 100 \text{ V/V}$$

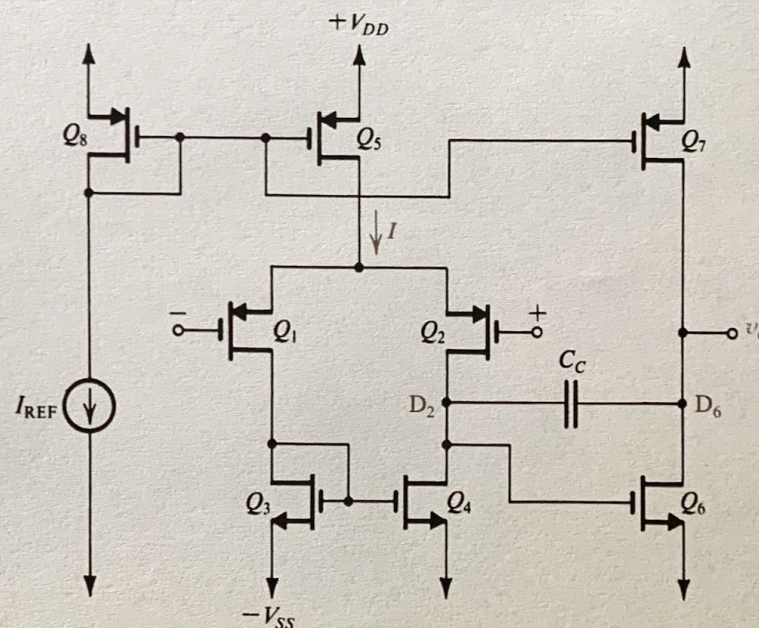


Example 9.6

Consider the circuit in Fig. 9.37 with the following device geometries (in μm).

Transistor	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8
W/L	10/0.4	10/0.4	2.5/0.4	2.5/0.4	20/0.4	5/0.4	20/0.4	20/0.4

Let $I_{\text{REF}} = 100 \mu\text{A}$, $V_{in} = 0.5 \text{ V}$, $V_{ip} = -0.5 \text{ V}$, $\mu_n C_{ox} = 400 \mu\text{A/V}^2$, $\mu_p C_{ox} = 100 \mu\text{A/V}^2$, $|V_A|$ (for all devices) $= 5 \text{ V}$, $V_{DD} = V_{SS} = 1 \text{ V}$. For all devices, evaluate I_D , $|V_{OV}|$, $|V_{GS}|$, g_m , and r_o . Also find A_1 , A_2 , the dc open-loop voltage gain, the input common-mode range, and the output voltage range. Neglect the effect of V_A on bias current.



见教材 P. 642